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Chem-biotech strategy to synthesize sustainable urethane-based polymers and additives from microbially produced hydroxyalkanoyloxy-alkanoates and polyol lipids

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ABSTRACT

Combining biotechnology and chemistry strategies seems to be one of the main drivers to develop innovative and sustainable macromolecular architectures for the next decades, starting from renewable carbon sources, such as biomass and plastic wastes. Herein, microbially produced biobased precursors, namely esterified 3-(3-hydroxyalkanoyloxy)alkanoates (HAAs), averagely comprised of ten carbons each fatty acid, and polyol lipids (PL) mainly structured with 3,5-dihydroxydecanoic acids connected to a mannitol head group as the largest fraction (78 %), have been used as building blocks to synthesize different urethane-based polymers and plasticizers as a proof of concept of this global approach. Polyurethanes (PU) were elaborated using HAAs and PL as chain extenders and then processed as films. The structure of these polyester-polyols significantly impacted the resulting properties. The hard segments (HS) on HAA-based PUs slightly increases from 40 to 46 % and the asymmetrical disposal of aliphatic chains hindered regular organization, making all samples fully amorphous. Conversely, the chain's entanglement benefited network cohesion. In contrast, chemically crosslinked PL-based PU can be processed as films with superior mechanical properties, despite having lower HS (33 %). Additionally, plasticizers based on HAAs, and PL significantly increased the flexibility and elasticity of polyvinyl chloride (PVC). These findings suggest that such sustainable polyester-polyols offer valuable potential for developing sustainable PUs, meeting the requirements of a circular (bio)economy.

1. Introduction

Associating biotechnology and chemistry (chem-biotech) is today one of the main driving forces behind the development of new sustainable, high-performance molecules and macromolecular architectures starting from different carbon resources (biomass or plastic wastes) [1–3]. These approaches take into account the beginning and the end of life of these materials as alternatives to reduce the dependence on finite fossil resources and mitigate the global environmental footprint and impact [4,5]. Over the past few decades, there has been a growing emphasis on developing renewable polymers, with a particular focus on polyurethane (PU) systems [6]. PU is a versatile polymer family mainly synthesized by polyaddition of polyols and isocyanates [7]. Representing the 6th most produced polymer family with around 5 % of worldwide production, most PUs goes to landfill [8]. From the 3.8 Mt produced in 2020 by the EU27+3, only 5 % was mechanically recovered, accounting for almost 25 % of plastic waste management [9]. Thus, considering the environmental impacts, various sustainable PUs formulations are been proposed [10,11]. Biomass, fatty acids and vegetable oils are among the most widely utilized renewable resources for producing biobased polyols for PU synthesis due to their cost-effectiveness and potential for large production [11–13]. However, aiming the combination of a more efficient and viable recycling process towards the principles of circular economy, significant efforts are being made to establish integrated chem-biotech recycling methods [14]. These methods involve the chemical deconstruction of polymeric matrices and bioconversion of the corresponding building blocks into innovative, value-added 2G polymeric materials with a wide range of

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macromolecular structures and unique properties [15–18]. In fact, taking the advantage of the metabolic pathways of microorganisms, bio-upcycling methods can turn into promising solutions by utilizing carbon-rich plastic waste streams as substrates in biotechnological processes [17,19,20].

During the past few years, suitable recycling and upcycling strategies based on a range of different mechanisms have demonstrated detachable efficiency using some plastic streams [21–26]. These include pyrolysis, glycolysis, alkaline hydrolysis, microwave-assisted, enzymatic approaches, and their combination to achieve superior bioconversion yield, especially for recalcitrant polymers [27–29]. Several microbial enzymatic approaches had demonstrated promising findings on the deconstruction of polymeric backbones possessing hydrolysable linkages, as the ester bonds being the core of most successful mechanisms, including polyester-based PUs [30–32], and many other plastic sources [33–40]. Bacterial co-culture comprising of *Bacillus subtilis* MZA-75 and *Pseudomonas aeruginosa* MZA-85 found to use polyester-based PU as a carbon source more efficiently, as compared to the individual strains [41]. Enriched mixed-microbial consortia using extracellular enzymes was found as efficient degrading cocktail against ester, urethane, ether, aromatic, and aliphatic chemicals groups [42]. Useful fungi species were also reported for their ability to degrade PU [43,44]. Recent studies have shown that microbial consortia was able to degrade a mixed recalcitrant polyethylene (PE), polypropylene (PP) [45,46], polyethylene terephthalate (PET), and polystyrene (PS) with PE [47] to generate a range of organic acids that were converted into a 2G products. To date, several sustainable monomers and value-added molecules have been obtained from chem-biotech strategies designed to specific plastic sources, but mostly from PET. These include diacids or vinyl-esters for synthesis of fiber-reinforced plastics [48], β-ketoadipic acid as a precursor for a nylon-6,6 from bis(2-hydroxyethyl) terephthalate (BHET) [29], catechol as coating agent [49], and the bioproduction of terephthalate acid (TPA) that turned into variety of compounds, including gallic acid, pyrogallol, muconic acid, and vanillic acid [28].

Recently, we have demonstrated a sequential conversion of post-consumer PET into hydroxyalkanoyloxy-alkanoates (HAAs) and further into sustainable poly(amide urethane) [50]. This process involves hydrolyzing PET with a thermostable polyester hydrolase and using the resulting ethylene glycol and terephthalic acid as feedstock for *P. umsongensis* GO16, which synthesizes the biopolyester polyhydroxyalkanoates PHA or HAAs (Fig. 1a). HAAs, is also found in rhamnolipid synthesis in *P. aeruginosa* PA01 [18], and are produced by genetically engineered *P. aeruginosa* PA01 in *P. putida* expressing the acyltransferase RhlA gene [51–53]. Besides PET monomers, HAAs can be derived from glucose, xylose, and ethanol, aiming for promising biotechnological applications [54,55] and as biosurfactants [53,56]. Analogously to HAAs, other bio-engineered and/or naturally occurring polyols may also display biotechnological advantages. This is likely the case for polyol lipids (PL), consisting of polyester oligo-dihydroxydecanoic acids (DDA) (formerly named as exophilin) connected to partially O-acetylated sugar alcohol head groups (arabitol, mannitol, threitol, and glycerol) (Fig. 1a) [57,58]. Secreted by *Aureobasidium pullulans* and known for its anticancer and antibacterial activities [59–61], PL have not yet been explored as renewable building blocks for sustainable PU synthesis despite their particular architecture.

Beyond the sustainable monomer/polymer loop, it is essential to consider the overall properties of the entire PU platform to ensure the production of feasible and reproducible materials from biobased building blocks. Recent findings from our group show that environmentally sustainable aromatic short diols from 4-hydroxybenzoic acid (HB.diol) and syringic acid (Syr.diol) affect phase segregation in thermoplastic PUs (TPUs) [62]. Aromaticity impacts hard segments (HS) domains significantly. Using 4,4′-methylene diphenyl diisocyanate (MDI, Fig. 1b) and aliphatic pentamethylene diisocyanate (PDI, Fig. 1b), HB.diol showed more phase segregation than conventional 1,4-benzenedimethanol and 1,4-butanediol (BDO, Fig. 1b). Syr.diol's asymmetry led to lower structuring and mechanical properties. In another recent study of our group, short linear poly-3-hydroxybutyrate (PHB) oligomers (oligoPHB-diol) were used to create TPUs with high biobased content [63]. Soft biobased dimeryl diisocyanate (DDI) was key in the soft segments (SS) domain, resulting in TPUs with varied properties, including high strength, ductility, and elastomeric behavior.

In this study, HAAs and polyol lipids (PL) have been developed and valued as renewable resources to synthesize innovative PUs and plasticizers, meeting the requirements of a circular (bio)economy. Beside our previous findings regarding bio-upcycling HAAs, herein we adopted the microbial pathway to produce HAAs from glucose, once our goal is to attend the proof of concept of using biobased polyols on a range of synthetic strategies to produce urethane-based polymers and additives. Meanwhile, PL were produced by the yeast-like fungus *A. pullulans* [64,65]. Polytetrahydrofurane (PTHF, Fig. 1b) was chosen as the soft segment in a two-step synthesis of PUs. The "macromolecular

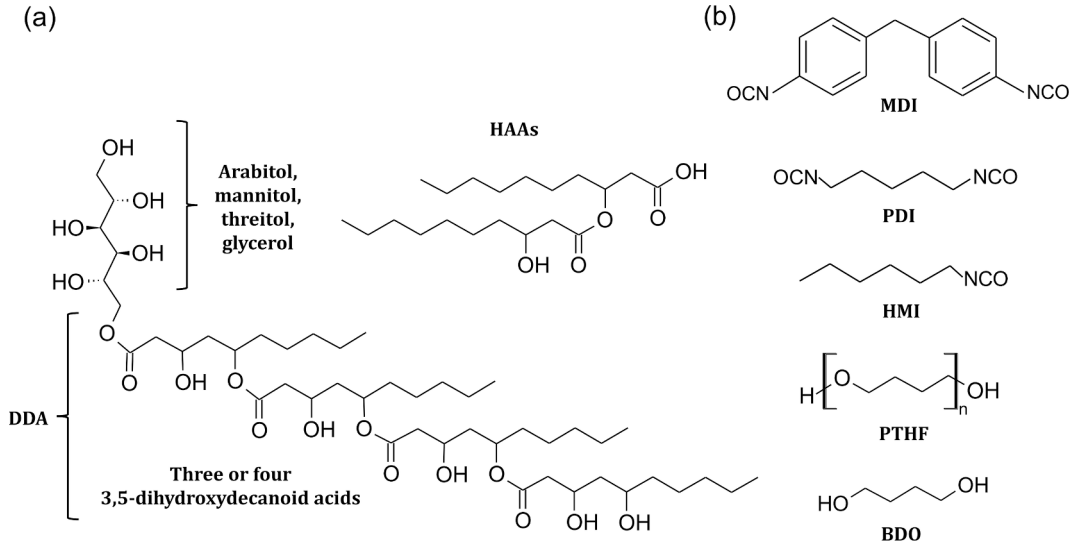

Fig. 1. Chemical structures of (a) a β-hydroxy fatty acids (HAAs) with chain lengths of ten carbon atoms in both fatty acids, and a polyol lipid comprising a polyol head group connected to an esterified oligo-dihydroxydecanoic acid (DDA) consisting of three or four 3,5-dihydroxydecanoic acids; (b) chemical structures of the different reactants used in the synthesis.

architecture-properties" relationships were accessed through the chemical structures of HAAs and PL, with varying ratios, PTHF molar masses, and the PDI (biobased) and MDI sorted as the diisocyanate of the given formulation. Then, the mechanical properties were accessed on those which could form films by hot-compression processing. Modified HAAs and PL structures were also evaluated as plasticizers into a classical thermoplastic, a polyvinyl chloride (PVC) matrix. Main behavior of these plasticizers and plasticized PVC can be found in the Supporting Information (SI).

2. Experimental

2.1. Chemicals

PTHF with a number-average molar mass of about 650, 1,000, and 2,000 $gmol^{-1}$ were acquired from BASF (Germany) under the trade name PolyTHF. MDI (Suprasec 1306) was purchased from Huntsman (USA). BDO (99 %) was purchased from Acros Organics (France). Tetrahydrofuran (THF, stabilized with 0.025 % BHT) and *N,N*-dimethylformamide (DMF, extra pure, stabilized with 0.025 % BHT) were purchased from Fisher scientific (France). PDI was kindly offered by Mitsui Chemicals Europe (Germany), under the trade name STABiO PDI. DMF, THF and PTHF were dried for 24 h, using 15 wt% of pre-activated molecular sieves 4 Å (molecular sieves activation: 50 °C in an oven under vacuum, 20 h). HAAs, PL, and BDO were dried under vacuum at 40 °C overnight prior to use. DMF-d7, $CDCl_3$, cholesterol (>99 %), and other common reagents and solvents were purchased from Sigma-Aldrich (France) and used as received.

2.2. Production of bio-based polyester-polyols

2.2.1. Hydroxyalkanoyloxy-alkanoates (HAAs)

HAAs were produced by using *P. putida* KT2440 KS3 [52] in a 2,500 L stirred tank bioreactor (Frings, Germany) with a filling volume of 600 L. To avoid any liquid foam formation, a combination of headspace aeration (25–30 L min^{-1}) and headspace overpressure (0.9 bar) was applied to ensure sufficient oxygen supply. Further details of the cultivation strategy were published by Weiser et al. [66]. The stirring frequency of the Rushton turbines was between 100 and 160 rpm. The pH was adjusted to 7.0 and controlled by adding NH_4OH (25 %). The cultivation temperature was controlled at 23 °C to reduce the oxygen demand over time. The fermenter was inoculated with 30 L to ensure an initial cell dry weight of 0.4 g/L ($OD_{600} = 0.3$).

The downstream process for HAA purification was conducted in the following way: First, a disc separation for biomass removal was carried out, followed by a crossflow filtration of the supernatant for further water removal. Next, discontinuous centrifugation of the retentate and a chromatographic step (with a C8 column and using methanol as eluent) were adjoined. As a final step, rotation evaporation was carried out to remove the methanol from the HAAs.

2.2.2. Polyol lipids (PLs)

Polyol lipids were produced by cultivation of *A. pullulans* 62031 in modified P medium [67] containing 5 g L^{-1} K_2HPO_4, 1 g L^{-1} NaCl, 0.6 g L^{-1} peptone, 0.4 g L^{-1} yeast extract, 120 g L^{-1} sucrose, and 0.4 g L^{-1} $MgSO_4 \times 7\ H_2O$. The cultivation was performed at 30 °C in Fernbach flasks with a filling volume of 500 mL and with horizontal shaking at 80 rpm for six days. Then, PL and cells were pelleted by centrifugation at 8,000 rpm for 15 min (Sorvall RC-5B Plus with A6.9 rotor) (Kendro Laboratory Products, Hanau, Germany). After discarding the supernatant, the pellet was resuspended in the same volume of acetonitrile to extract PL. After overnight incubation at room temperature, cells were removed from the acetonitrile by centrifugation (as previously performed) and subsequent filtration with a bottle-top sterile filter (polyether sulfone film with 0.2 μm pore size) (St. Louis, Missouri, USA). Subsequently, the acetonitrile was evaporated under a constant airflow to obtain the PL.

2.3. Synthesis of PUs

Different approaches were performed to use HAAs and PL as building blocks for PU-based materials. Both sustainable building blocks were used in a two-step process [17,68]. In a first step, a prepolymer possessing an isocyanate (NCO) ending chains was produced, leading to a molar ratio with the polyol's hydroxyls, [NCO]/[OH], equal to 2. Subsequently, a high molar mass polymer is obtained by introducing a chain extender in the form of a short diol. The resulting [NCO]/[OH] is equal to 1.05 and a short diol was added after 40 min of prepolymerization. The reactions were performed in a three-necked round bottom flask under argon flow and mechanical stirring for 2 h at 75 °C. The MDI was melted beforehand and added to the flask. In case of HAAs, a first group of products was designed by reacting PTHF 1,000 g mol^{-1} with MDI during the prepolymerization, then HAAs and/or BDO, previously diluted in DMF were added as chain extenders at different nOH proportions. Hereafter, in a second group, HAAs and/or PTHF at different nOH proportions were diluted in DMF and reacted with MDI during the prepolymerization. BDO was added as chain extender. In sequence, prepolymers based on MDI, biobased PDI and PTHF at variable molar masses, *i.e.* 650, 1,000 and 2,000 g mol^{-1}, were prepared followed by the addition of PL as chain extender. Since the presence of multiple reactive sites of PL would lead to an unavoidable chemical crosslinked network, the reactants were chosen to explore the feasibility and processability of the final PU with thermoset properties. The content of hard segments (%HS) was calculated as described in previous reports [68,69]. All products were left overnight at room temperature for curing and then dried for 24 h at 50 °C in vaccum oven. PU films were then processed with around 0.2 mm thickness.

2.4. Characterization techniques

Infrared spectra were recorded by a Fourier transformed infrared spectrometer Nicolet 380 (Thermo Electron Corporation) working in reflection mode and equipped with an ATR diamond module (FTIR-ATR). The IR spectra were acquired in the range of 4000–400 cm^{-1} by accumulating at least 32 scans with resolution of 4 cm^{-1}. The degree of phase separation (DPS) was calculated using the relative deconvoluted area of vibrational bands ranging in 1680–1715 cm^{-1} corresponding to H-bonded HS, over the sum with the bands ranging in 1715–1735 cm^{-1}. The experimental area was obtained by fitting Voigt functions using PeakFit™ (v. 4.12) software for peak deconvolution processing.

High-resolution ^1H and ^{31}P NMR spectra were recorded on a Bruker 400 MHz spectrophotometer. About 100 mg of HAAs and 15–20 mg of soluble PU samples were dissolved in 400 μL of $CDCl_3$ ($\delta_H = 7.26$ ppm) for ^1H NMR spectroscopy. The ^{31}P NMR spectroscopy was used to quantify the hydroxyl content of the polyols. About 15 to 20 mg of HAAs was derivatized with 2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholane in pyridine/$CDCl_3$ (1.6:1 v/v) in the presence of cholesterol ($\delta_P = 144.48$ ppm) as internal standard, and quantified as described in a standard protocol [70,71]. The number of scans was set to 16 for ^1H and 128 for ^{31}P nuclei, while 15 s was set for relaxation delay. The acquisition ^1H and ^{31}P NMR spectra of PL followed the same protocol, although being dissolved in DMF-d7 ($\delta_H = 8.10$ ppm).

The average molar masses (M_n, M_w) and polydispersities (Đ) of soluble PU samples were determined by Size Exclusion Chromatography (SEC) coupled with an Acquity APC apparatus from Waters in dried THF at 40 °C and flow rate of 0.6 mL min^{-1}. About 10 mL containing 3–4 mg of dissolved polymer solution were filtered with 0.2 mm PTFE filters and injected into three connected columns (Acquity APC XT 450 Å 2.5 μm 4.6 × 150 mm, 200 and 45). The calibration was obtained using polystyrene (PS) standards. Detection is performed with PDA-UV detector at 254 nm. The final PU samples which were not fully soluble in THF or in other conventional solvents for SEC, due to higher brittleness and/or chemical crosslinking, could not have their molar masses determined.

Thermal degradation was studied by thermogravimetric analyses

(TGA). Measurements were conducted under dry nitrogen (flow rate = 25 mL min^{-1}) using a Hi-Res TGA Q5000 apparatus from TA Instruments. The samples (1–3 mg placed in an aluminum pan) were heated up to 750 °C at 10 °C min^{-1}.

Differential scanning calorimetry (DSC) was performed using a TA Instrument Q200. Samples (2–3 mg) were placed in sealed aluminum pans and analyzed under nitrogen flow (50 mL min^{-1}). A three-step procedure with a 10 °C min^{-1} ramp was applied as follows: (1) heating up from room temperature to 185 °C and holding for 3 min to erase the thermal history, (2) cooling down to -90 °C and holding for 3 min, (3) heating (second heating) from −90 °C to 200 °C.

Dynamic mechanical analyses (DMA) were analyzed using a TA Instruments Discovery hybrid Rheometer HR-3 in dynamic mode. The rheometer was equipped with rectangular torsion geometry. Temperature range varied from -75 °C to 150 °C at 2 °C min^{-1}, at 1 Hz of spinning frequency and 0.01 % of strain.

Uniaxial tensile tests were performed on an Instron 5567H (USA) with a 10 kN load cell. The experiments were carried out at room temperature with a constant crosshead speed of 20 mm min^{-1}. Sets of five dumbbell-shaped samples with dimensions of approximately 30 × 5 × 1 mm^3 were cut and tested. Young's modulus, tensile strength at break (σ_{max}) and elongation at break (ε_{max}) were calculated from the curves.

3. Results and discussion

3.1. Analysis of the bioproduction of HAAs

Since HAA production by *P. putida* leads to strong liquid foam formation in the case of submerged aeration of the fermentation broth, headspace aeration was applied. Consequently, oxygen transfer was limited to a maximal value of 0.2 mmol L^{-1} h^{-1} due to the small gas–liquid surface between headspace and culture broth (Fig. 2a). As indicated by the online signal of the dissolved oxygen tension, oxygen was limited in the liquid phase after approx. 8.5 h of cultivation time. Therefore, biomass formation as well as HAAs formation were linear until the end of the process (Fig. 2b). The maximum HAAs concentration was reached after 48 h with a value of 1.6 g L^{-1}.

3.2. Analysis of the bioproduction of polyol lipids (PL)

Polyol lipid production by *A. pullulans* NRRL 62031 was carried out using 120 g L^{-1} sucrose. Although not shown in the current data,

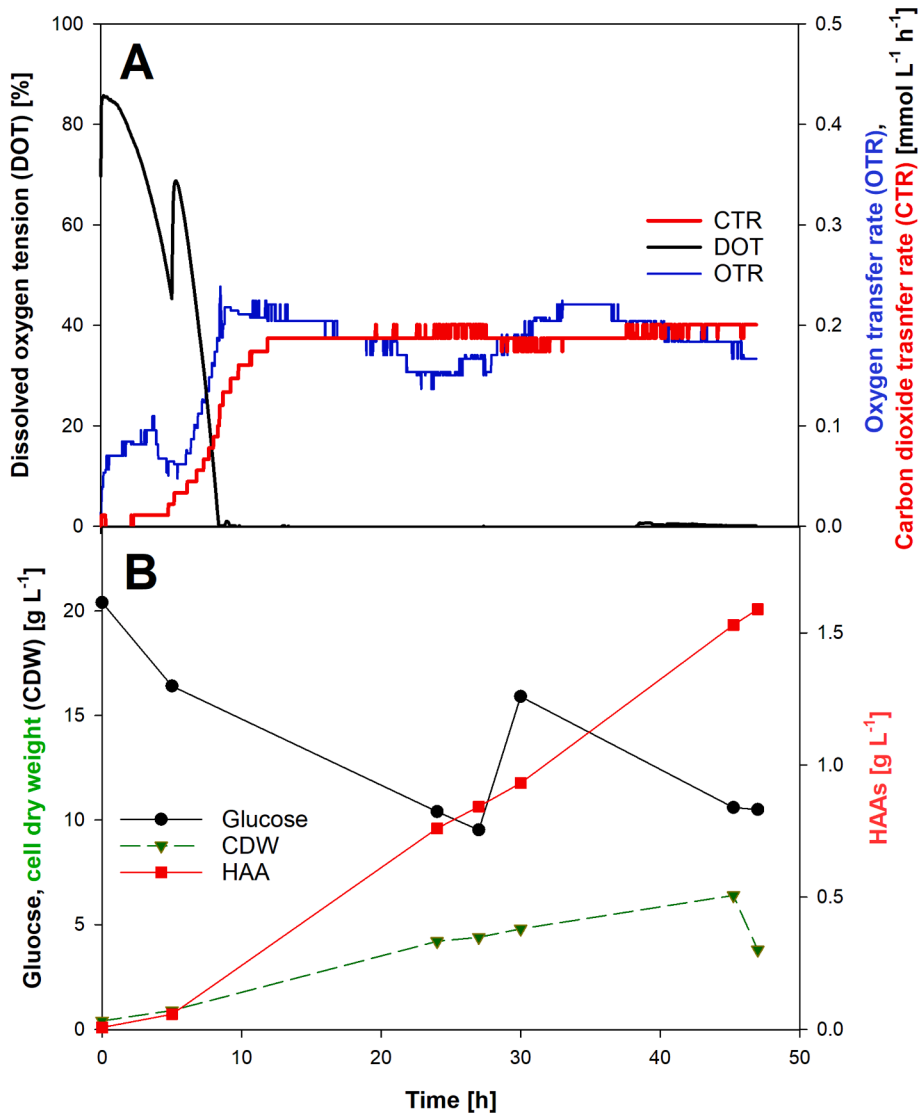

Fig. 2. Batch culture of *P. putida* KT2440 KS3 with glucose pulse at 26 h; (a) Online signals of oxygen transfer rate, carbon dioxide transfer rate, and dissolved oxygen tension; (b) Offline measured concentrations for glucose, cell dry weight, and HAAs; Culture conditions: Reactor volume: 2,500 L, filling volume: 600 L, pH controlled: 7, headspace aeration rate: 25–30 L min^{-1}, temperature: 21–23 °C, stirring rate: 100–160 rpm, headspace overpressure: 900 mbar.

previous studies indicate that the culture's growth typically ceases after approximately 75 h, which aligns with the observed trend towards a plateau in biomass accumulation and the courses of PL concentration and pH value (Fig. 3a). PL production commenced after the lag phase and continued to increase linearly throughout the cultivation period, achieving a maximum concentration of 1.2 g L^{-1} at 75 h. Concurrently, the pH of the medium decreased sharply during the first 50 h and then stabilized around a value of 3.6. This drop in pH is likely due to the production of acidic by-products during sucrose metabolisation.

Fig. 3b illustrates the distribution of PL congeners in the mixture (after 150 h), which is similar to values reported before [65]. As can be seen, the mannitol lipids make up the largest fraction (78 %), while around 18 % of the molecules lack polyols as head group, hence being named only as DDA [72]. The relative share of acetylated and non-acetylated PL congeners is not conserved with the head group of the biosurfactant. The data show a high proportion of acetylated congeners in the Man-C10-C10-C10 group compared to the other congeners. In all the shake flasks combined 10 g of the polyol lipid were produced.

3.3. Analysis of the elaborated PU

3.3.1. Influence of synthetic strategies on the physical aspect and chemical structure

The choice of reaction pathway using HAAs and PL as PU building blocks significantly impacted the feasibility and reproducibility of the resulting material. The confirmation of HAAs structure and the quantification of hydroxyl content were conducted using ^{1}H and ^{31}P NMR. The detailed spectrum assignments can be found in SI (Figures S1a and S1c). Eight different MDI-HAA PUs were synthesized (Figure S2a). Due to the high brittleness of the product synthesized under variable HAAs ratios during prepolymerization, we focused on those prepared under variable HAAs content as chain extenders. This group ensured flexible PU films. Although PTHF's primary OH groups allow faster reactions, the OH groups of HAAs compete in presence of MDI during prepolymerization. It is worth noticing that such trend relies on the fact that the shorter and branched HAA/ PTHF competition to the MDI reactive sites occur at the almost same beginning. This is the main reason for achieving low range coherence over the matrix, leading to inhomogeneous and brittle products. The physical entanglement and high asymmetry hinder HS organization. Therefore, processable PU films were only obtained using HAAs as chain extenders.

The main resulting products, named MDI-HAA followed by the average number of HAAs wt% relative to BDO, were characterized by ^{1}H NMR spectroscopy. As shown in Fig. 4, all MDI-HAA architectures exhibited high-intensity signals at 3.58 and 1.66 ppm corresponding to O–CH and C–CH residues of the PTHF structure, while the multiplet at 7.3–7.1 ppm refers to the C=CH of MDI [73,74]. Superimposed signals in the 1.27–1.77 ppm region were assigned to methylene protons from BDO and HAAs' congeners [68]. The successful linkage of HAAs to the prepolymer resulted in a downfield shifted signal with two peaks in the 9.8 to 10.6 ppm range, corresponding to secondary –NH groups. These peaks confirmed that both urethane and amide linkages were properly formed regardless of the HAAs content [50]. Distinct functionalization is a consequence of HAAs' reactive sites, such as secondary hydroxyls and carboxylic acid. Other signals mostly refer to the HAAs structure (Fig. 4).

Similarly, the structure confirmation and quantification of hydroxyl groups of PL were conducted using ^{1}H NMR (Figure S1b in SI) and ^{31}P NMR (Figure S1d in SI), respectively. The PL-based PUs synthesized with MDI using PTHF 1,000 g mol^{-1} and 2,000 g mol^{-1} led to inhomogeneous and brittle products, mainly due to the crosslinking nature of PL, and to the hard segments from aromatic residues The relatively flexible PDI molecule also resulted in a highly brittle product when reacting with PTHF 650 g mol^{-1} (Figure S2b in SI). Once we have found brittle products after carrying the polymerization even with a longer and intrinsically more flexible PTHF chains (1,000 and 2,000 g mol^{-1}), using a shorter PTHF (650 g mol^{-1}) to produce MDI-PL would lead to brittle products as well. Only PDI-PL (PTHF 2,000 g mol^{-1}) has been found to be sticky. As the hard segment content is insufficient or poorly distributed, the soft segment properties (low strength, tackiness) dominate. This imbalance led to a polymer that is cohesive but lacks the tensile properties needed for film formation. Therefore, at 2,000 g mol^{-1}, PTHF introduces a relatively high level of chain mobility and contributes to viscosity, further reinforcing the sticky nature. Finally, the route carried with PTHF 1,000 g mol^{-1} and PDI was the only one that yielded a flexible and homogeneous PU film. As expected, due to its crosslinking nature, the PDI-PL (PTHF 1,000 g mol^{-1}) product was completely insoluble in common organic solvents. Despite the multiple reactive sites of PL leading to a chemically crosslinked network, the resulting PDI-PL does not behave like a typical thermoset, as it can be processed by hot-compression molding. Therefore, the following characterizations of PDI-PL pertain only to its thermal and mechanical properties.

Table 1 presents the M_n and Đ values of MDI-HAA-based PUs determined from SEC chromatograms (Figure S3 in SI). A decreasing trend in polymeric chain lengths is observed as the HAAs content increases. This behavior is likely due to the lower reactivity of carboxylic acid and secondary –OH groups (about three times less) compared to the

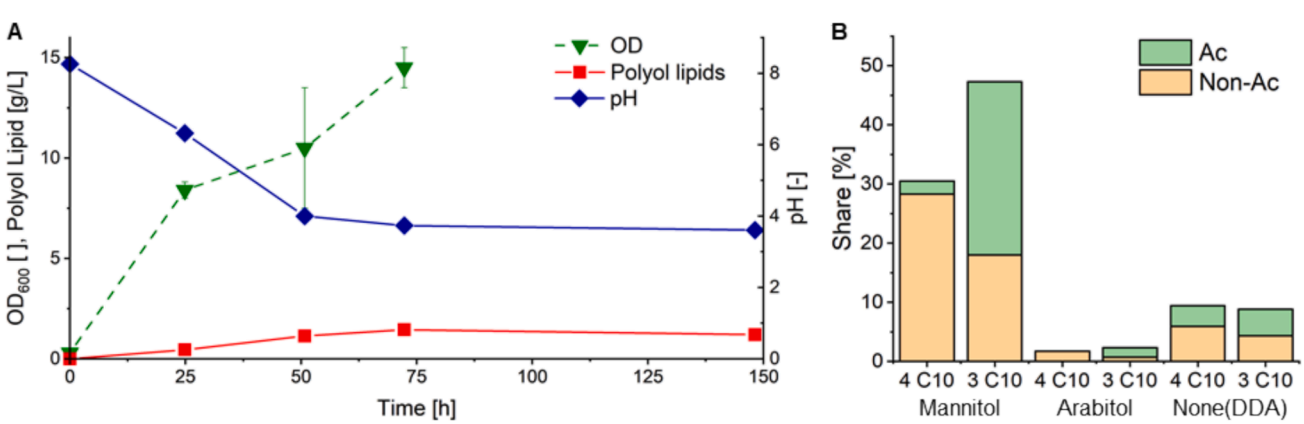

Fig. 3. Flask culture of *A. pullulans* NRRL 62031 in modified P medium with 120 g/L sucrose to produce polyol lipids; (a) The OD_{600} (green triangles, dashed line) indicates the optical density at 600 nm, which is a measure of cell growth. PL concentration (red squares, solid line) is measured in gL^{-1}. The pH of the culture medium (blue diamonds, solid line) is plotted on the secondary y-axis. Error bars represent standard deviations from biological duplicates; (b) Distribution of PL congeners. The relative share (%) of acetylated (Ac, green) and non-acetylated (Non-Ac, orange) polyol lipid congeners is shown. The labels "4 C10″ and "3 C10″ refer to the number of di-hydroxy-decanoic acid groups incorporated in the PL. The head group is indicated below. "None" means that the congeners lack a headgroup and are therefore a DDA. Therefore, for practical reasons, we adopted PL to refer to the resulting mixed product. Cultivation conditions: Temperature: 30 °C, Fernbach flasks, filling volume: 500 mL, horizontal shaking at 80 rpm for six days.

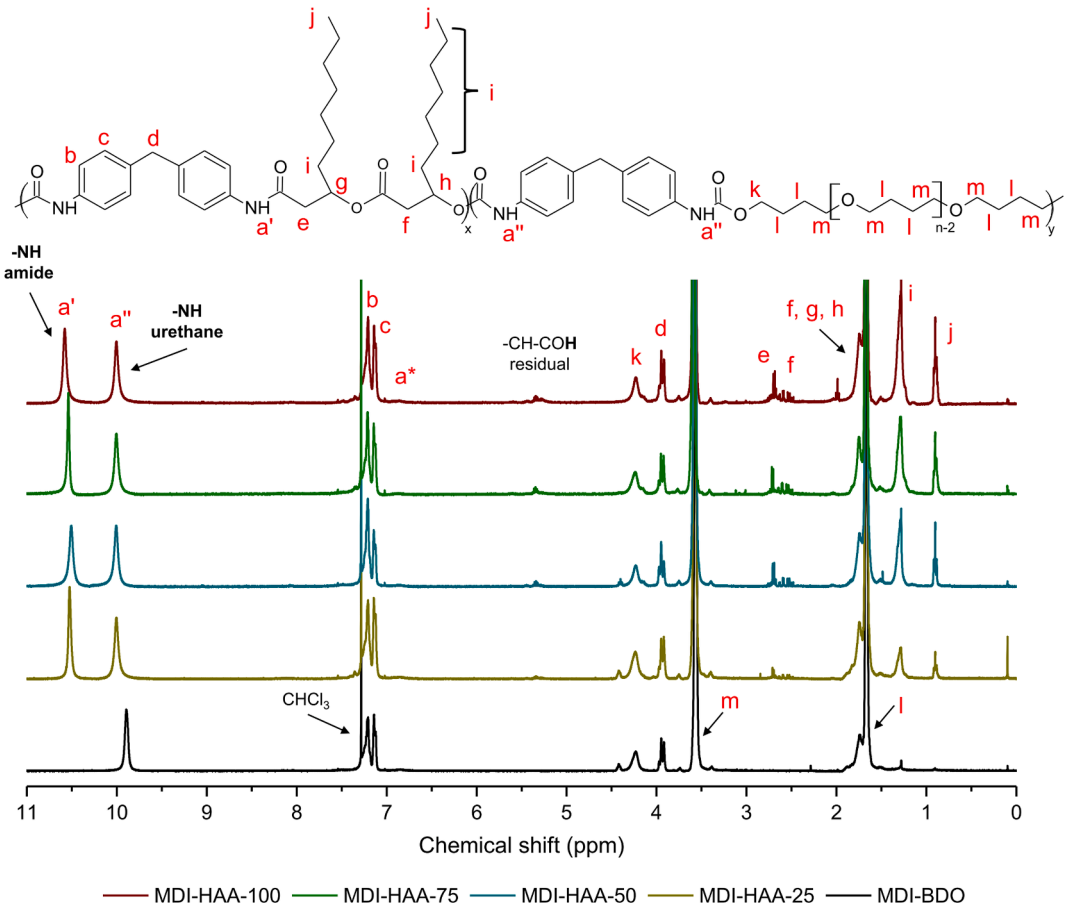

Fig. 4. ^{1}H NMR spectra of MDI-HAA-based PUs and the reference MDI-BDO in $CDCl_3$ and the assignment of the main signals.

Table 1
Relative content of the building blocks, HS, M_n, Đ of the synthesized PUs.

PU materials	Diisocyanate (wt%)	Long polyol (wt%)	Chain extender (wt%)	HS (wt %)	M_n (kg/mol)	Đ
MDI-BDO	32	63	5	37	69.3	2.3
MDI-HAA-25	31	60	9	40	53.1	2.7
MDI-HAA-50	30	58	12	42	42.4	2.5
MDI-HAA-75	29	56	15	44	34.9	2.9
MDI-HAA-100	28	54	18	46	38.8	3.4
PDI-PL	21	67	11	33	–	–

primary –OH groups of BDO, leading to shorter chain lengths [75]. Additionally, the steric hindrance caused by the HAA's branched structure hinders effective access to the reactive sites. The motion restrictions of HAAs play a significant role in the formation of HS, which slightly increases with higher HAAs content (Table 1), as the aliphatic congeners contribute to a network with more entangled segments. Branched configurations have a reduced hydrodynamic size as compared to linear chains of equivalent molar mass [76].

3.3.2. Thermal properties

The thermal stability of MDI-HAA PUs was evaluated by TGA (Fig. 5). The corresponding $T_{95\%}$ and maximum degradation rates (T_{deg}) values are listed in Table 2. All PUs exhibited high $T_{95\%}$, indicating successful formation of thermally resistant materials. However, the presence of HAAs made the resulting PUs less thermally stable, consistently showing lower $T_{95\%}$ as HAAs content increases. This weight loss trend is directly related to the increase in HS segments [76]. All MDI-HAA PUs exhibited lower thermal stability compared to the reference MDI-BDO, which comprises higher content of SS domains. This reduced thermal stability is due to the initiation of thermal degradation in PUs through the cleavage of urethane (and amide) linkages, primarily comprised in the HS domains [77]. This trend also aligns with the corresponding M_n, as higher chain lengths typically lead to higher thermal stability. Three to four mains T_{deg} steps, ranging from 150–210 °C, 210–330 °C, and 330–450 °C, can be assigned to HAA-based formulations. Their dependences on HAA content are unclear.

In contrast, the fully aliphatic architecture of PDI-PL does not provide higher values $T_{95\%}$ as compared to other non-aromatic based PUs from previous studies [62,78]. The relatively low first degradation step (T_{deg1}, Table 2) is likely due to the high asymmetry and lack of HS organization [79]. However, the second and third degradation rates (T_{deg2} and T_{deg3}) are considerably higher for PDI-PL as compared to MDI-HAA PUs, due to the higher SS content (lower HS) [80]. In all cases, the remaining dissociated isocyanates and hydroxyls, typically assigned between T_{deg2} and T_{deg3}, can form thermally stable urea, carbodiimides, and isocyanurate rings, which are completely degraded after T_{deg3} [81].

The corresponding thermal properties of PUs were evaluated by DSC and presented in Table 2. In all cases, the lower crystallinity of HS segments is clearly evidenced by the weakly defined melting temperature (T_m), during the heating step, and absence of the crystallization temperature (T_c), during the cooling step (Figure S4b in SI), as compared to the reference. These behaviors confirm the low HS organization, as a direct consequence of HAAs' branched chains that would considerably prevent the regular packing of the chains. Such trend was also found in methyl side groups of other chains extenders [82,83]. Since MDI-HAA

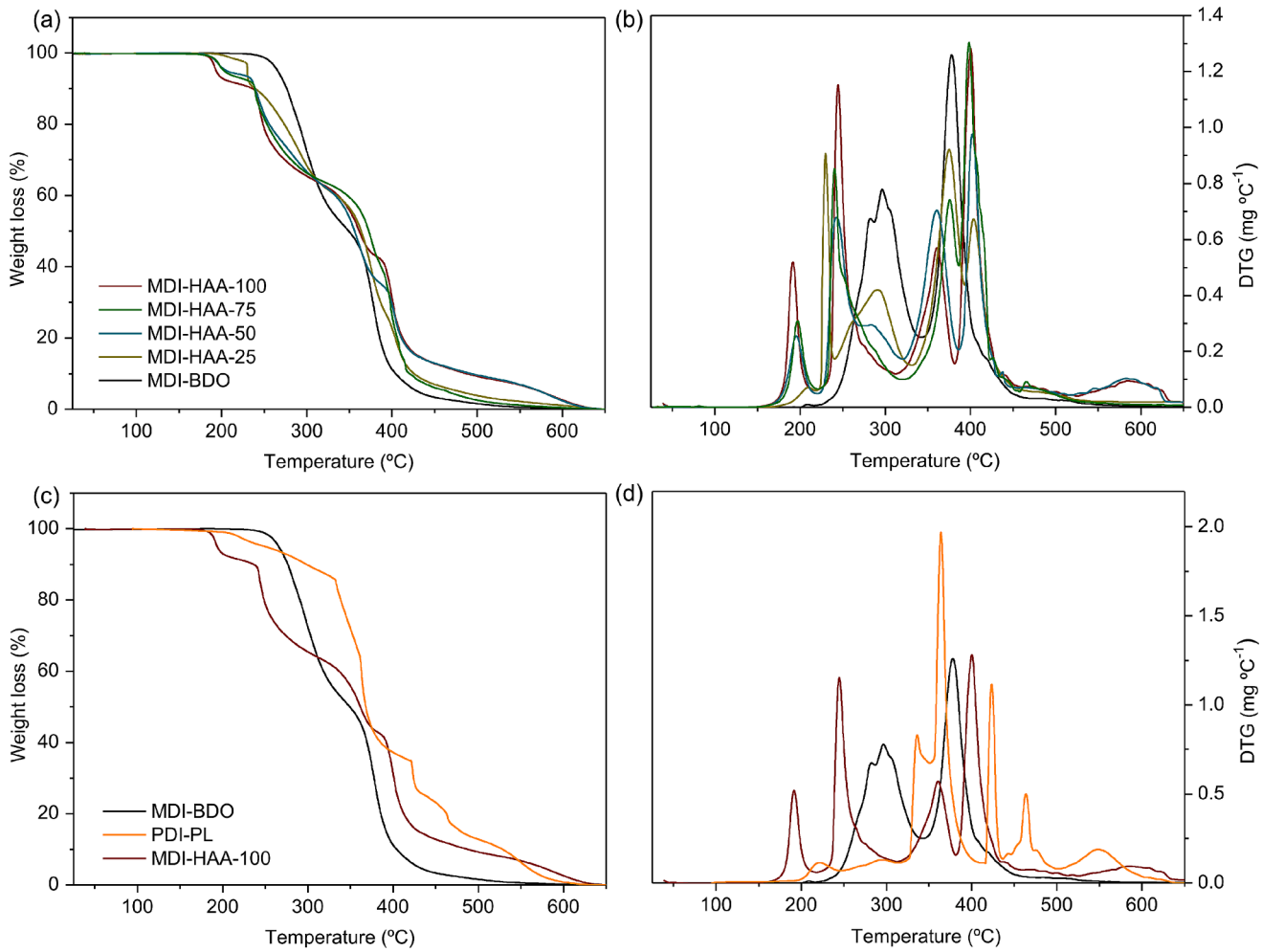

Fig. 5. TGA curves of MDI-HAA (a) and PDI-PL and MDI-HAA-100 (c) with respect to the reference MDI-BDO; DTG curves of PU (b, d).

Table 2
DPS values and thermal events of synthetized PU, calculated from FTIR spectra and obtained from TGA and DSC analyses, respectively.

PU materials			TGA						DSC	
	HS (wt%)	DPS (%)	$T_{95\%}$(°C)	T_{deg1}(°C)	T_{deg2}(°C)	T_{deg3}(°C)	T_{deg4}(°C)	–	T_g(°C)	T_m(°C)
MDI-BDO	37	72	275	296	377	487	–		−38	130
MDI-HAA-25	40	46	230	229	290	374	403		−43	129
MDI-HAA-50	42	46	207	198	243	360	403		−44	134
MDI-HAA-75	44	43	203	196	239	375	398		−45	137
MDI-HAA-100	46	38	194	192	244	359	400		−47	136
PDI-PL	33	54	203	219	364	422	464		−51	–

and PDI-PL PUs are extensively amorphous, it is expected absence of reorganization and microphase rearrangement during the heating step, thus indicating good segregation of SS [74]. As similarly found elsewhere, the symmetry and steric hindrance of the monomers pertaining the HS domains are decisive to define the composition and crystallization of these domains [84,85].

The amorphous behavior makes the crystallizable fraction minimal, or too small imperfect crystallites able to exhibit a distinct melting transition. Indeed, the microstructure could involve phase mixing or disrupted crystalline domains due to the presence of hydrogen bonding or chain rigidity, further diminishing the possibility of a sharp melting peak. If the underlying thermal event is inherently broad or ill-defined, it would still not yield a clear T_m. These features make it challenging to isolate a distinct T_m, regardless of DSC setting parameters. The absence of a distinct melting peak for hard segments in polyurethane systems is well-documented in the literature [86,87]. This phenomenon is primarily attributed to the microphase-separated morphology and the amorphous nature of the hard segments, which disrupts their crystallization.

All PUs exhibited a glass transition temperature (T_g) below −20 °C, consistent with the T_g of SS domains. The T_g does not significantly increase with higher HS content, indicating that HS domains are well separated from the SS ones [88]. The chemical crosslinking in PDI-PL contributes to the amorphous nature of PDI-PL.

3.3.3. FTIR-ATR spectroscopy

The FTIR spectra of all synthesized PUs confirm their chemical structure and provide key information on hydrogen bonding related to phase organization [85]. The complete spectra and characteristic H-bonded carbonyl regions from NCO-terminated architectures are

displayed in Fig. 6. The successful consumption of prepolymer is indicated by the absence of the NCO stretching band at 2260 cm^{-1}. The attachment of MDI and PDI is verified by the broad N–H stretching band assigned between 3500 and 3200 cm^{-1}. MDI-HAA PUs showed variations in CH_3 and CH_2 asymmetric stretching vibrations at 2930 and 2850 cm^{-1}, and in C–H, C–C, and C–O bending vibrations across the fingerprint region, without a straight relationship with HAAs' content (Fig. 6a). In contrast, at this same interval, PDI-PL exhibited more overlapped vibrational bands, suggesting lower phase organization as compared to MDI-based PUs (Fig. 6c).

Significant variations in H-bonded urethane groups remain crucial for understanding overall structure and morphology. As shown in Fig. 6b and Fig. 6d, the vibrational band around 1730–1720 cm^{-1} corresponds to non-H-bonded C=O groups, originally from ester groups of HAAs and PL, overlapping with non-H-bonded urethane groups linking the SS domains [84,89,90]. The band around 1700 cm^{-1} represents strongly H-bonded urethane groups, with an increase in intensity at 1685 cm^{-1} indicating even stronger bonding, similar to what is typically observed in MDI-BDO systems [91]. The intensity of the 1700–1685 cm^{-1} region increases proportionally with DPS values, thus decreasing with HAA content (Fig. 6b and Table 2). This trend indicates a higher proportion of non-H-bonded urethanes compared to disordered H-bonded ones [92], confirming that as the content of HAAs' congeners increases, the HS organization is proportionally reduced due to the lack of H-bonding of the C=O groups from amide and urethane linkages.

Although strong H-bonded urethane groups are formed in case of PDI-PL, a higher content of non-H-bonded groups remains (Fig. 6d). This behavior aligns with the expressive number of reactive sites on PL, which sustain a significant quantity of non-reacted C=O groups across the PDI-PL network. As consequence, higher levels of SS domains (lower HS, Table 1) and lack of HS organization are found, as compared to MDI-based PUs.

The FTIR results are consistent with the thermal behavior observed through DSC thermograms. The high degree of amorphous character in MDI-HAA and PDI-PL PUs is straightly related to the lower levels of H-bonding, indicating reduced segregation of HS domains. In contrast, MDI-BDO exhibited measurable T_c, thus higher crystallinity due to stronger H-bonded urethane groups. This confirms the low compatibility between HS and SS domains, and good phase segregation.

3.3.4. Thermomechanical and mechanical properties

Dynamic Mechanical Analysis (DMA) was performed on PU films to study the evolution of modulus and determine the relaxation temperature (T_α) corresponding to the maximum tanδ peak. The storage modulus (G') and loss factor tanδ dependence versus temperature are shown in Fig. 7. Since the rheological profiles of MDI-HAA PUs were considerably similar, Table 3 only lists T_α values for MDI-HAA-50 and MDI-HAA-100, as well as MDI-BDO and PDI-PL for comparison purposes.

As observed in Fig. 7a and 7b, G' of MDI-HAA drops sharply after

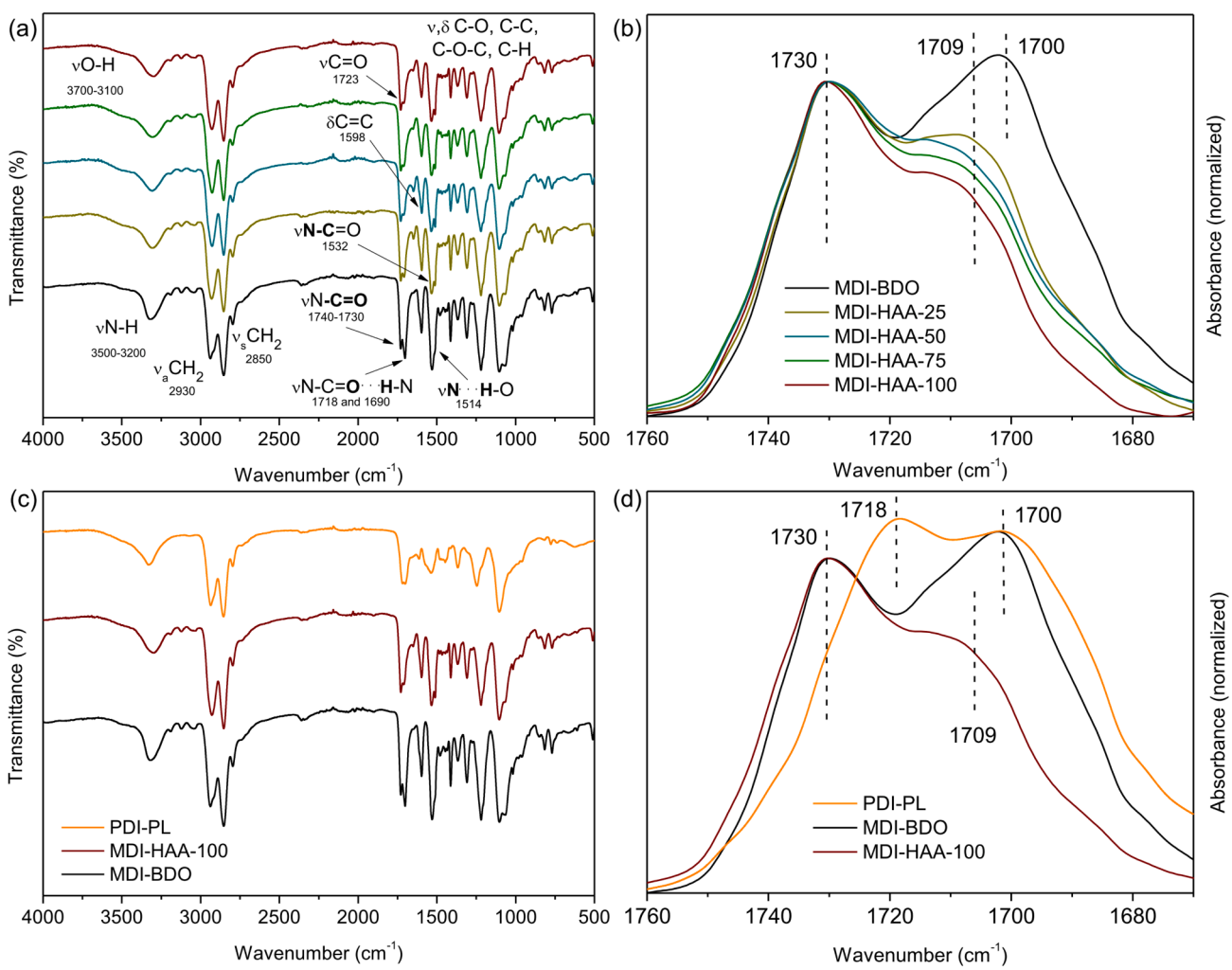

Fig. 6. FTIR spectra of MDI-HAA (a) and PDI-PL and MDI-HAA-100 (c) with respect to the reference MDI-BDO; normalized by the absorbance of carbonyl stretching region (b, d).

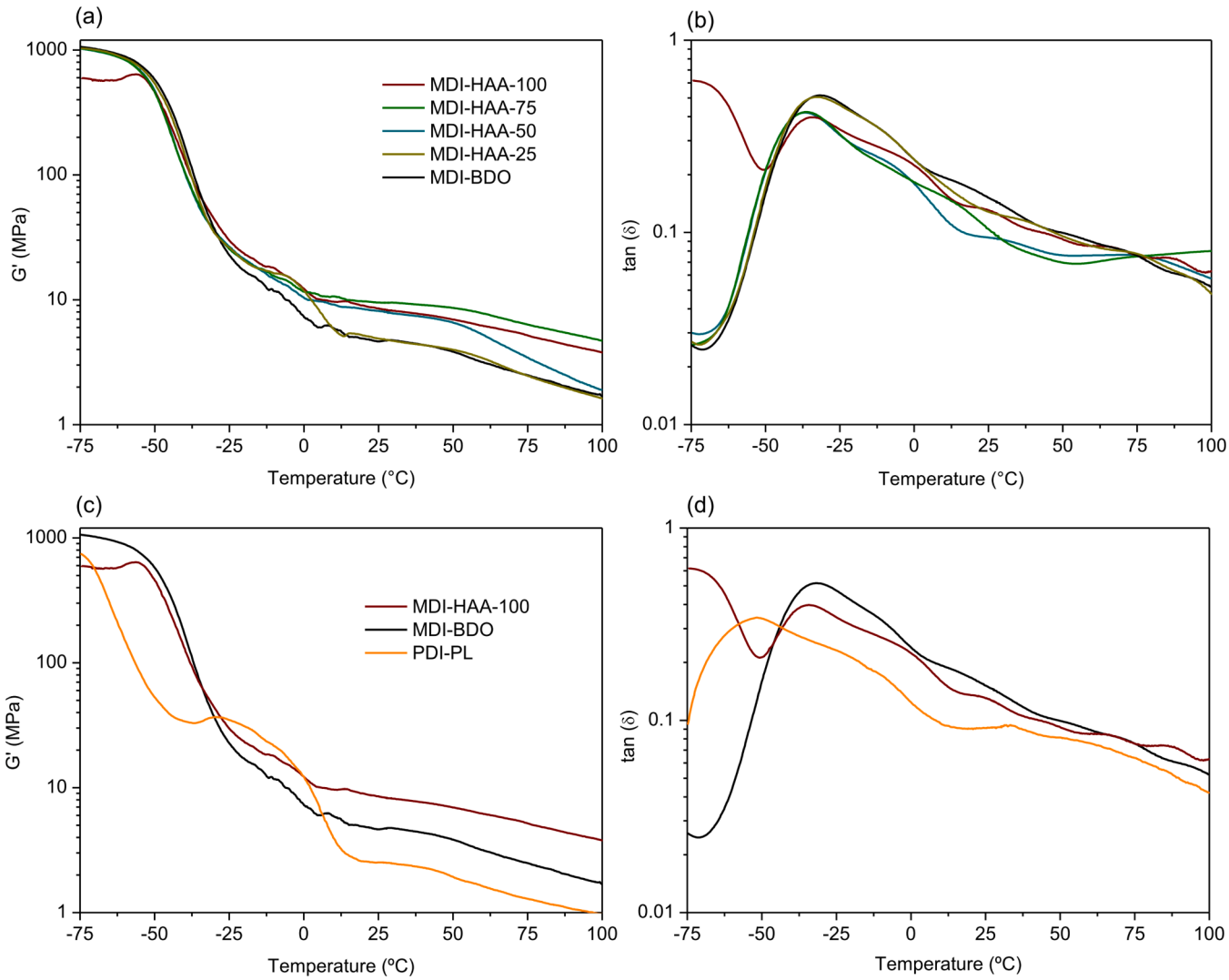

Fig. 7. Storage modulus G' of MDI-HAA (a) and PDI-PL and MDI-HAA-100 (c) with respect to the reference MDI-BDO; loss factor tan δ of these PU films (b, d), acquired by DMA.

Table 3
Thermal, thermomechanical, and mechanical results on the synthetized PU films, from DMA analysis and uniaxial tensile tests.

PU materials	DSC	DMA	Uniaxial tensile tests		
	T_g(°C)	T_α(°C)	E (MPa)	σ_{max}(MPa)	ε_{break}(%)
MDI-BDO	−38	−35.1	12.8 ± 4.1	29.1 ± 7.0	503 ± 50
MDI-HAA-50	−44	−31.4	6.5 ± 3.4	21.3 ± 3.8	374 ± 61
MDI-HAA-100	−47	−33.7	4.8 ± 2.0	13.6 ± 4.2	241 ± 45
PDI-PL	−51	−67.7	7.5 ± 4.8	17.1 ± 5.3	410 ± 47

−50 °C, displaying a rubbery plateau typical of thermoplastic elastomers that extends over a wide temperature range [88]. For MDI-based PUs, T_α values closely align with the T_g of SS domains. While the relationship between T_α and HS was less evident for MDI-HAA, important insights can be inferred when compared with PDI-PL (Fig. 7c and Fig. 7d). The lower T_g and T_α values of PDI-PL compared to other MDI-based PUs result from the combined features of lower rigidity of PDI and chemical crosslinking of PDI-PL, which prevent high HS content from dispersing into SS domains and promote earlier elastomeric behavior during temperature screening. Consistently with previous findings, higher T_α values suggest a weaker crosslinked network in HS domains [62,74]. Hence, PDI-PL exhibits better phase segregation than MDI-HAA, thus being consistent with DPS values.

The rubbery plateau at low temperatures showed some G' variation among MDI-based samples (Fig. 7a). The stronger H-bonding in MDI-BDO leads to a higher number of physical crosslinking sites, resulting in higher G' values [88,93]. However, it is noteworthy that the network arrangement of MDI-HAAs is favored by the entanglements of HAAs' congeners, despite weaker H-bonding from urethane groups. Even though the HS domains are not crystalline, they behave like physical crosslinks, resulting in a higher plateau modulus. As the HAAs content increases, the amount of HS increases, leading to higher G' in the plateau interval. Conversely, the G' of PDI-PL starts to drop at a lower temperature than MDI-based PUs and reaches the lowest G' values right after a shoulder-shaped inflection interval (−37 to 12 °C). This inflection indicates the presence of chemical crosslinks, which contribute to the highest elastomeric behavior. At higher temperatures, above this inflection interval, the thermal energy increases the mobility of the PDI-PL flexible segments, resulting in a noticeable drop in G′. This is not a reflection of crosslink network breakdown but rather the softening of the polymer's soft segment. In contrast, the dissociation of hydrogen bonds in MDI-HAA leads to a significant and irreversible decrease in G′. While

both PDI-PL and MDI-HAA experience a drop in G′ with increasing temperature, the extent and mechanism differ.

The mechanical properties of the main samples were evaluated using uniaxial tensile tests. The nominal stress–strain curves are shown in Fig. 8, and the average values of Young's modulus, tensile strength, and elongation at break are provided in Table 3. MDI-HAA-100, MDI-HAA-50, and PDI-PL exhibited shorter elastic zones and lower moduli compared to MDI-BDO. As expected, within the MDI-HAA PUs, higher HAAs content led to lower mechanical properties, primarily due to the decrease in M_n values relative to MDI-BDO. This behavior also reflects the lower rigidity of HAAs' structure whether compared to unbranched or aromatic chain extenders [94].

The presence of HAAs in the macromolecular architectures provided significant flexibility. This flexibility may be attributed to the low – but still present – H-bonding, as analogously found for ether groups [95], and to the effective entanglement of congeners acting as physical crosslinks. However, the lack of long-range ordering and undetectable crystalline domains results in a lower Young's modulus.

Finally, the presence of chemical crosslinks enables PDI-PL to achieve extensive deformation without breaking. This behavior is advantageous, as PDI-PL demonstrates a useful deformation profile, not requiring rigid monomers, such as aromatic building blocks in the architecture, to achieve this behavior.

3.4. Analysis of the performance of the synthesized sustainable plasticizers

An additional study has been based on the development of plasticizers, which is the first additive used by the thermoplastic industry, particularly focused on PVC-based materials. The main results concerning the synthesis of plasticizers and the performances of plasticized PVC films have been given in SI. Sustainable plasticizers have been pre-modified with PDI and HMI to ensure higher compatibility with the PVC matrix. All approaches resulted in a decrease in the T_g, with respect to the one recorded for neat PVC, confirming the plasticization effect. In contrast to the findings from HAA-based plasticizers, the mechanical tests performed for HMI-PL revealed significantly higher flexibility and deformation ability, mainly due to the higher free volume added to the PVC matrix in response to increased content of hexyl-branched structures.

4. Conclusion

Distinct renewable biobased building blocks were successfully produced using bioengineered microbes, aiming the synthesis of sustainable PUs and plasticizers of PVC. For PU synthesis, two-step pathways were set for HAAs/BDO and PL as chain extenders, using the diisocyanates MDI and PDI and the polyol PTHF 1,000 g mol^{-1}, to produce flexible and processable materials. This strategy based on bulk polymerization was further implemented using PDI to produce HAA-based plasticizers and adapted to a one-pot direct synthesis, aiming at the synthesis of HMI-based plasticizers.

The macromolecular architectures and the corresponding behaviors of these PU architectures were significantly impacted by these chain extenders. Despite the slight HS increase with HAAs content, DPS goes to an opposite trend, suggesting that the HAA congeners hinder HS packing in a regular way. Simultaneously, those congeners favored network cohesion once the entanglement of its aliphatic branched structure can display the role of physical crosslinking. As demonstrated, T_g (SS) was averagely constant even with HS variation, indicating that the HAAs' content causes a negligible impact over the separation of HS and SS domains in MDI-HAA PUs. Such an amorphous trend also meets the findings for PDI-PL PU. Once leading to unavoidable chemical crosslinking, this material was surprisingly processable by hot-compression molding partially due to the absence of aromatic residues and intrinsic higher flexibility of PDI as compared to MDI. Such configuration resulted in lower T_g and HS, however, higher DPS and averagely higher Young's modulus and elasticity as compared to HAA-based PUs.

Following, the study on developing sustainable plasticizers showed that pre-modifying HAAs and PL with PDI and HMI enhances PVC compatibility. All approaches confirmed the effective plasticization, as showed by lower T_g values. Unlike HAA-based plasticizers, HMI-PL showed greater flexibility and deformation due to increased hexyl-branched structures.

The overall outcomes achieved by the PU-based materials produced in this study enabled assessing novel relationships within the expected architectures and the resulting behaviors. In this sense, the selection of each building block and synthesis pathways alongside HAAs and PL was of utmost importance in face of the feasibility and performance of the final products. These findings paved the way for emerging opportunities, possibly expanding the usage of HAAs and polyol lipids for different industrial PU applications after a scaling-up step. In this context, different formulations could be tested to fit with a large range of applications such as is civil and automotive engineering, two of the main polymer markets where sustainable and innovative materials are increasingly necessary to meet the circular (bio)economy demands.

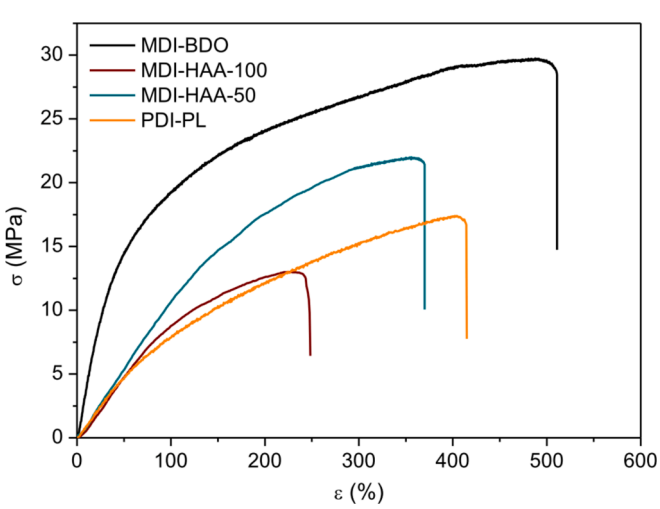

Fig. 8. Representative stress–strain curves of MDI-HAA, PDI-PL and MDI-BDO, acquired from uniaxial tensile tests.

CRediT authorship contribution statement

William M. Facchinatto: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Agathe Mouren:** Methodology. **Gina Welsing:** Writing – original draft, Methodology. **Karsten Willing:** Methodology. **Lars Regestein:** Writing – original draft. **Till Tiso:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Lars M. Blank:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Luc Avérous:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Luc Averous reports financial support, administrative support, and equipment, drugs, or supplies were provided by CNRS Délégation Centre-Est. Luc Averous reports administrative support and equipment, drugs, or supplies were provided by University of Strasbourg. Lars Blank reports financial support, administrative support, and equipment, drugs, or supplies were provided by RWTH Aachen University. If there are other authors, they declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper].

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113621.

Data availability

Data will be made available on request.

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